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## What is a Module?
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**** A module is a Python file that contains reusable code.**

A module can contain:

Variables
Functions
Classes
Statements

Module file extension is .py

Ex:

calculator.py

student.py

backend.py

fronend.py

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## Why Modules?
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Modules are used for:

Code reusability
Code organization
Reducing duplicate code
Easy maintenance
Splitting large programs into small files
Using built-in Python features

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## Simple Meaning
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**** Module means one Python file.**

**** If we write functions in one Python file and use them in another Python file, that file is called a module.**

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## Types of Modules
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Python has mainly 3 types of modules:

- 1) Built-in modules
- 2) User Defined Modules
- 3) External Modules / Third Party Modules

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## Built-in Modules
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**** Built-in modules are already available in Python.**

**** We can use them directly by importing.**

Examples:

- 1) math
- 2) random
- 3) datetime
- 4) os
- 5) sys
- 6) json

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##### Example of built-in modules #####
```

```
import keyword
import sys
import math
import random
import datetime
import os

print(keyword.kwlist)

a = 20
print(type(a), sys.getsizeof(a))

print(math.sqrt(25))
print(math.pow(2,3))
print(math.ceil(10.2))
print(math.floor(10.2))
print(math.factorial(5))
print(math.pi)

print(random.randint(1,10))
print(random.randint(100000,999999))

today = datetime.date.today()
now = datetime.datetime.now()

print("Today Date : ", today)
print("Today Date & Time : ", now)

print(os.getcwd())

print(sys.version)

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## External Modules
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**** External modules are not available by default.**

**** We need to install them using pip.**

Note: pip is a package manager for Python.

Examples:

- 1) numpy
- 2) pandas
- 3) matplotlib
- 4) requests
- 5) fast api
- 6) django

how to install a module

Ex : pip install matplotlib

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## Create simple plot and graph using matplotlib
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```

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
y = [10, 20, 30, 40, 50]
```

```
plt.plot(x, y)
```

```
plt.title("Simple Line Chart")
plt.xlabel("X Values")
plt.ylabel("Y Values")
```

```
plt.show()
```

```
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```

```
import matplotlib.pyplot as plt
```

```
students = ["Ravi", "Anil", "Priya", "Sneha"]
marks = [75, 85, 90, 80]
```

```
plt.bar(students, marks)
```

```
plt.title("Student Marks")
plt.xlabel("Student Names")
plt.ylabel("Marks")
```

```
plt.show()
```

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## import Keyword
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**** import keyword** is used to use one module inside another Python file.

syntax :

```
import module_name
```

Example :

```
import math
import sys
import os
```

Modules with alias name

Note: We can give another name to module using "as" keyword.

Ex :

```
import math as m
```

```
print(m.sqrt(5))
```

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=====
## from import
=====
```

**** from import** is used to import specific function or variable from a module.

Syntax :

```
from module_name import function_name
```

```
from module_name import function_name1, function_name, function_name3
```

Example ::

```
from math import sqrt, factorial
```

```
print(sqrt(5))
print(factorial(5))
```

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## Create User-defined Module
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```

Step-1 :: Create a file named "calculator.py"

```
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calculator.py
```

```
def add(a, b):
    return a + b
```

```
def sub(a, b):
    return a - b
```

```
def mul(a, b):
```

```
        return a * b

def div(a, b):
    return a / b
-----

## Step-2 : ## Use User-defined Module in "main.py" file

-----
# import calculator
# import calculator as calc

from calculator import add, mul

print(add(10, 20))
print(mul(10,20))
-----
```